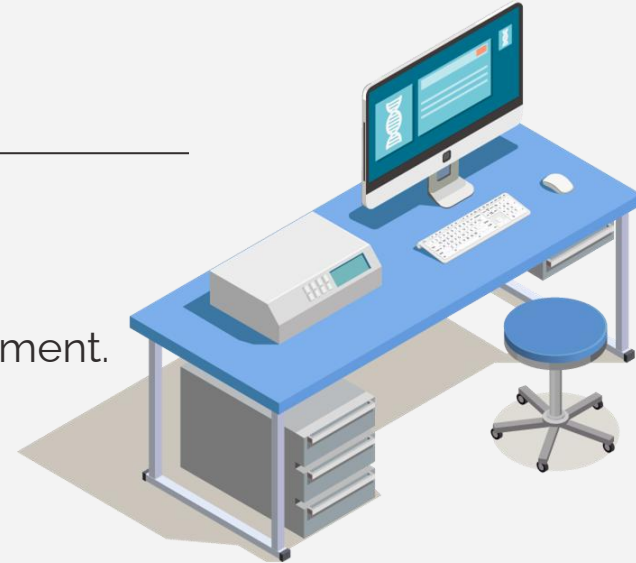




Biomedical Data Acquisition Course (**BMD351**)



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Associate Professor,
Biomedical Informatics Department.



02

Biomedical Signals



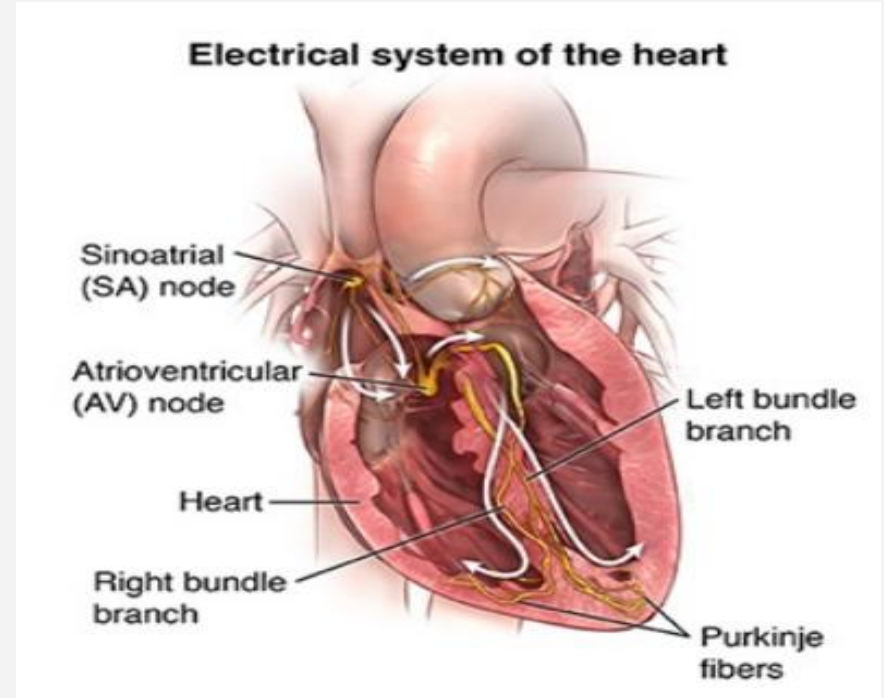
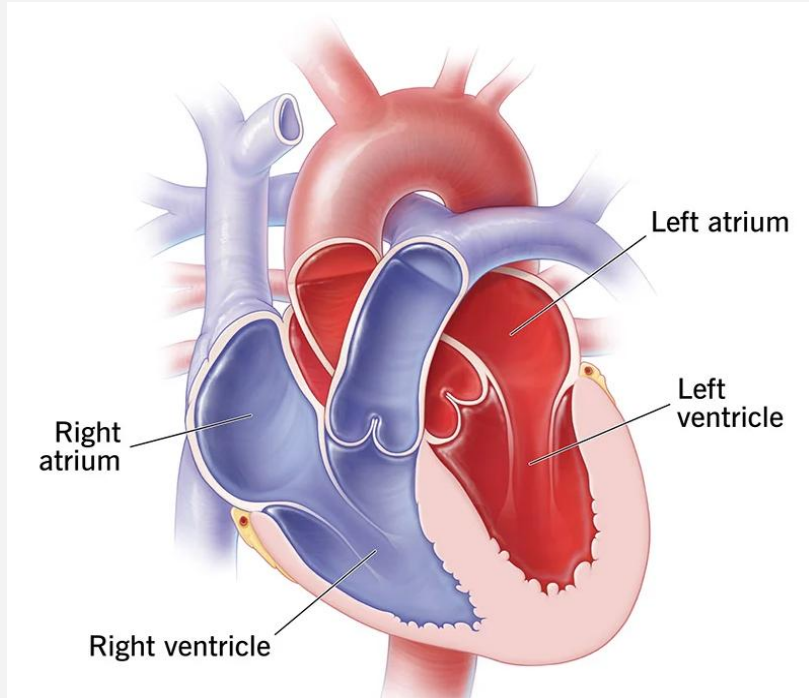
ECG / EKG

Heart Signals

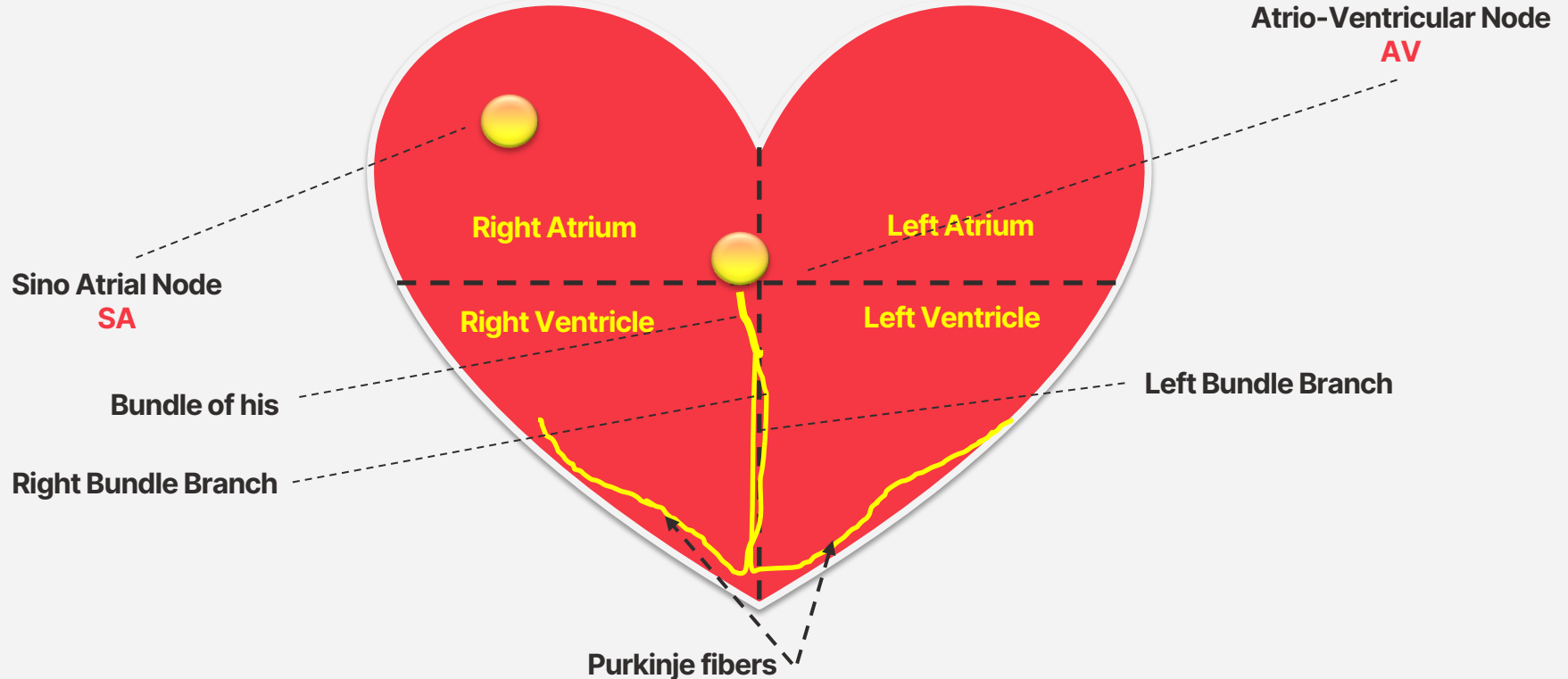
Bio-Electric Signals

- Electrical potential generated by the cell membrane potential called action potential and mainly generated by nerve and cell muscle.
- Examples:
 - (1) Electroencephalography (EEG): brain signal recordings.
 - (2) Electromyography (EMG): muscle signal recordings.
 - (3) Electrooculography (EOG): eye signal recordings.
 - (4) Electrocardiography (ECG/EKG): heart signal recordings.

Electrical System of the heart



Electrical System of the heart



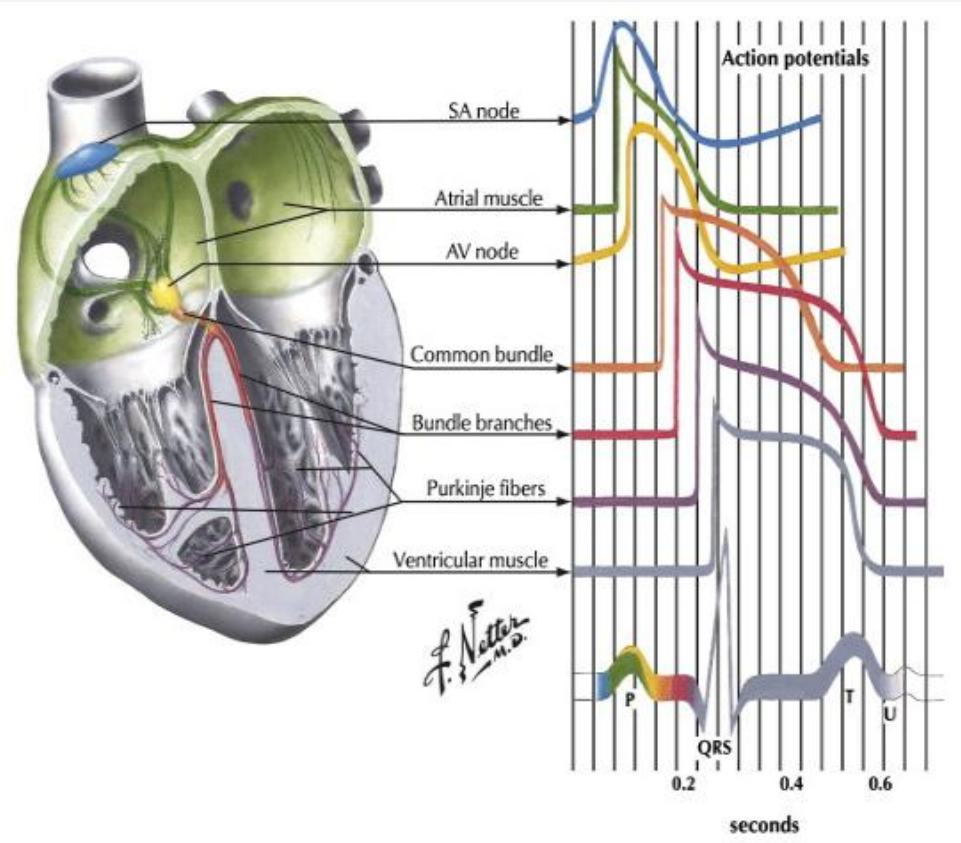
Electrical System of the heart

- During each heartbeat, a healthy heart has an orderly progression of depolarization that starts with pacemaker cells in the sinoatrial node, spreads throughout the atrium, and passes through the atrioventricular node down into the bundle of His and into the Purkinje fibers, spreading down and to the left throughout the ventricles. This orderly pattern of depolarization gives rise to the characteristic ECG tracing.

Electrical System of the heart

- Sino Atrial Node (SA Node) exists within right Atrium and represents the Natural Pacemaker of the heart, this is where the beginning of the conduction takes.
- From the SA node the conduction goes to the AtrioVentricular node (AV Node) which exists within the border of right atrium and right ventricle.
- From AV node it goes to single structure called the bundle of his and will go into two separate cells (Left and Right bundle branches) and eventually Purkinje fibers.

Source	Heart Rate
SA	60-100 Beats per Minute
AV	40 – 60
Bundle of His	40 – 60
left and right bundle branches to Purkinje fibers	40 -20



Electrocardiography

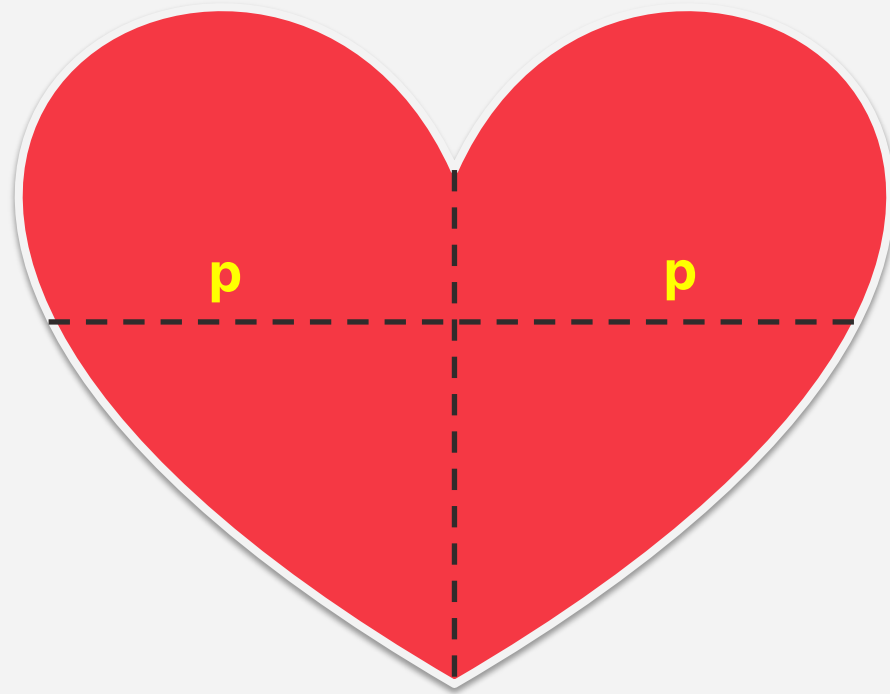
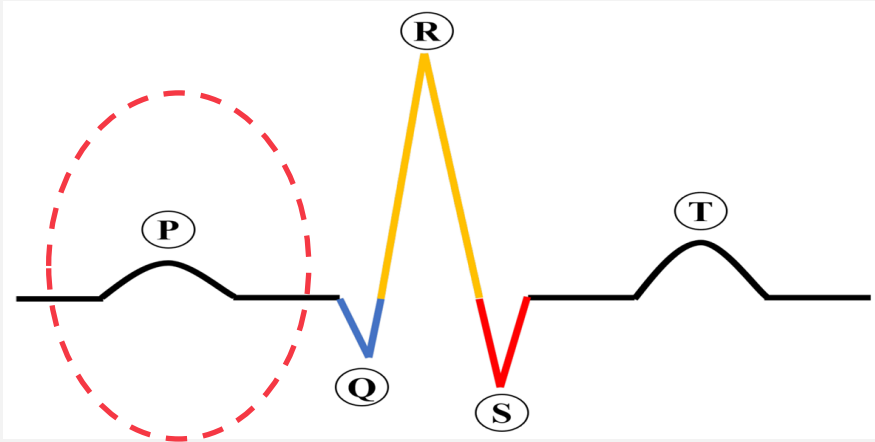
- **Electrocardiography (ECG/EKG)** is the process of producing an electrocardiogram, a recording of the heart's electrical activity through repeated cardiac cycles.
- The resulting signal is called the electrocardiogram and the instrument is called electrocardiograph.
- The electrocardiogram (ECG or EKG) is a depiction of the electrical activity of the heart in the form of a graph of voltage versus time as measured from electrodes placed on the surface of the body.
- These electrodes detect the small electrical changes that are a consequence of cardiac muscle depolarization followed by repolarization during each cardiac cycle (heartbeat).
- Changes in the normal ECG pattern occur in numerous cardiac abnormalities, including cardiac rhythm disturbances

Electrocardiography

- Each ECG complex represents a single heart beat and is comprised of different waves, originally named by Willem Einthoven, who won the Nobel Prize in 1928 for his pioneering work on electrocardiography.
- Normal rhythm (main components to an ECG) produces three entities – a P wave, a QRS complex, and a T wave – that each have a fairly unique pattern.
 - The P wave represents atrial depolarization.
 - The QRS complex represents ventricular depolarization.
 - The T wave represents ventricular repolarization.

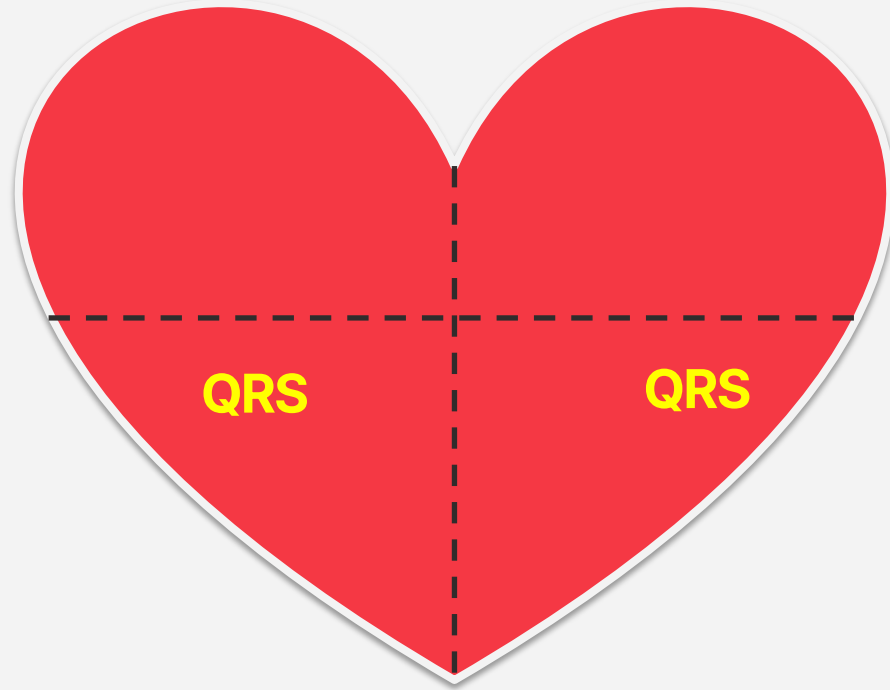
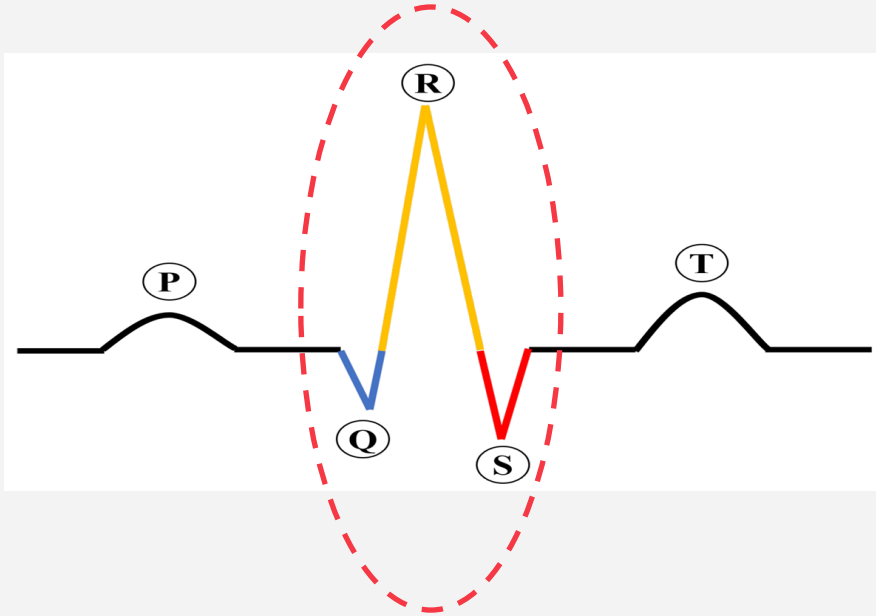
P Wave

Atrium depolarization



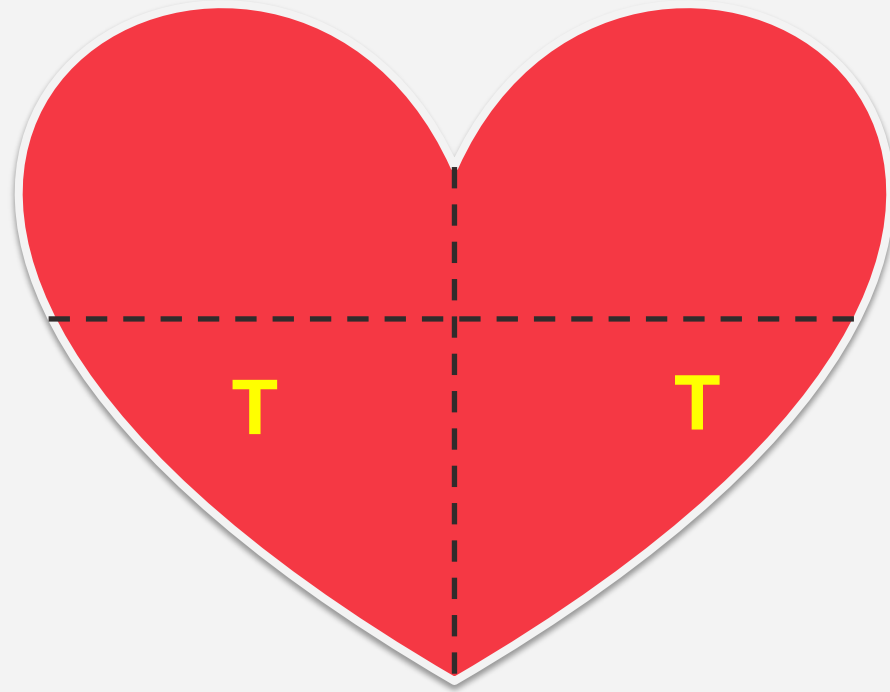
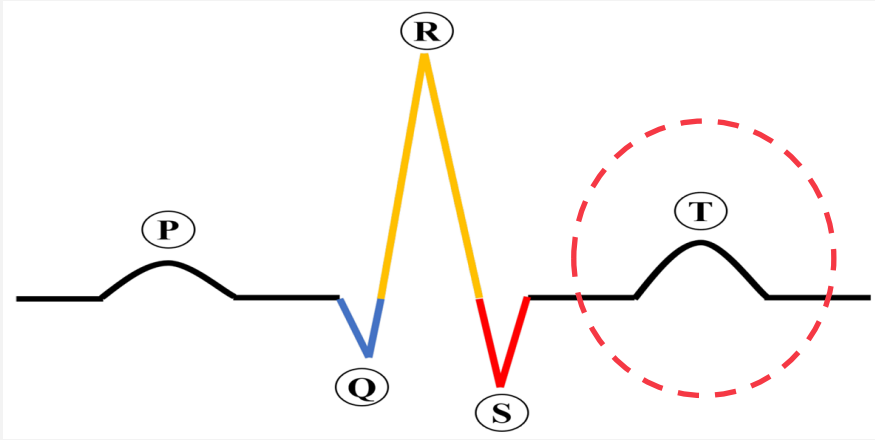
QRS Complex

Ventricle depolarization

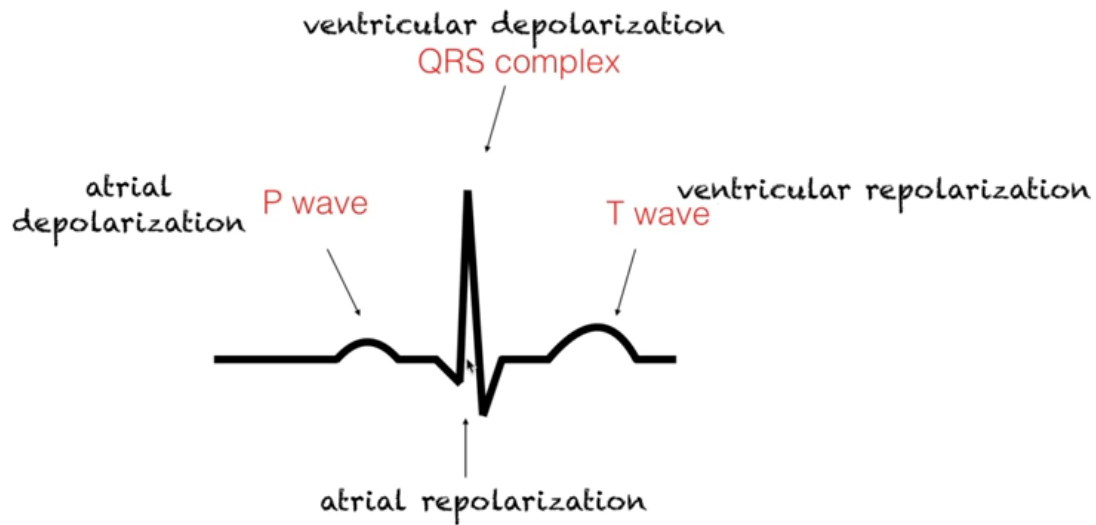


QRS Complex

Ventricle repolarization

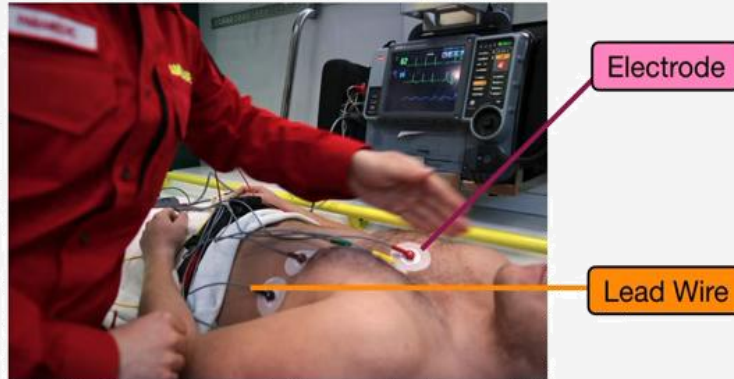


ECG Waves



Electrocardiography

- In a conventional 12-lead ECG, ten electrodes are placed on the patient's limbs and on the surface of the chest.
- The overall magnitude of the heart's electrical potential is then measured from twelve different angles ("leads") and is recorded over a period of time (usually ten seconds).
- In this way, the overall magnitude and direction of the heart's electrical depolarization is captured at each moment throughout the cardiac cycle.



12 Lead ECG Placement

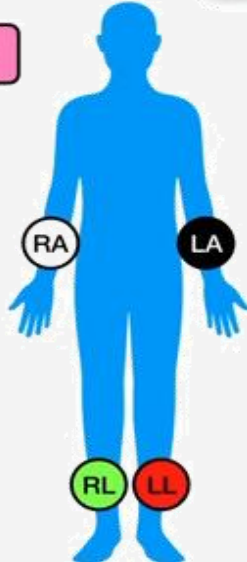
Types of Electrodes

Limb Electrodes

There are 4 electrodes placed on the Limbs:

Lead Wires

Right Arm (white)
Left Arm (black)
Right Leg (green)
Left Leg (red)

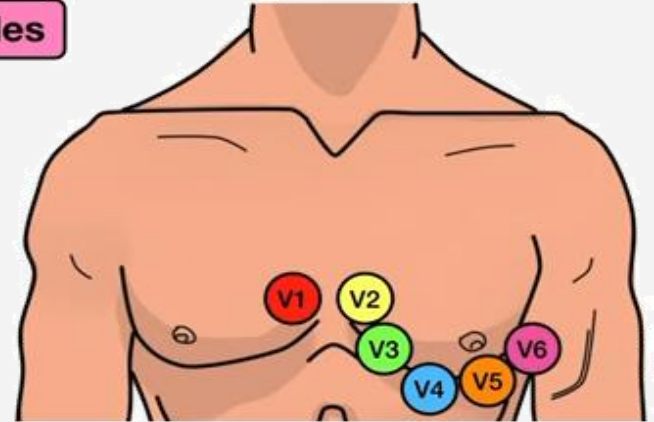


Precordial Electrodes

There are 6 electrodes placed on the Chest:

Lead Wires

V1 (red)
V2 (yellow)
V3 (green)
V4 (blue)
V5 (orange)
V6 (purple)



Electrocardiography

- **White on the right** = Place electrode on the right arm typically above the right wrist
- **Clouds over grass** = (clouds = white over grass (green)) green lead wire on the right leg above the right ankle
- **Smoke over Fire** = Smoke = black lead on the left arm over fire (red) on the left leg.
- **Ride Your Green Bike On Pavement** for Precordial electrodes and lead wires.

"Ride Your Green Bike On Pavement"

Ride = Red = V1

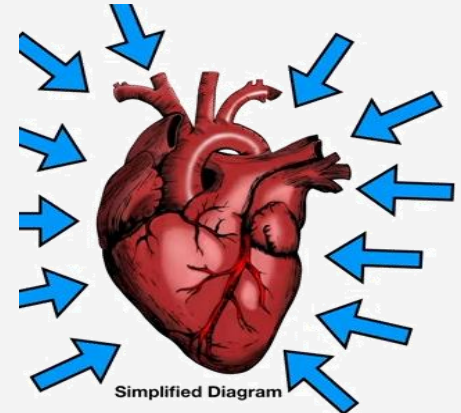
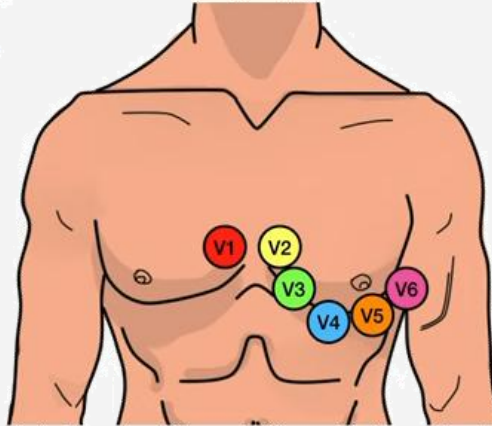
Your = Yellow = V2

Green = Green = V3

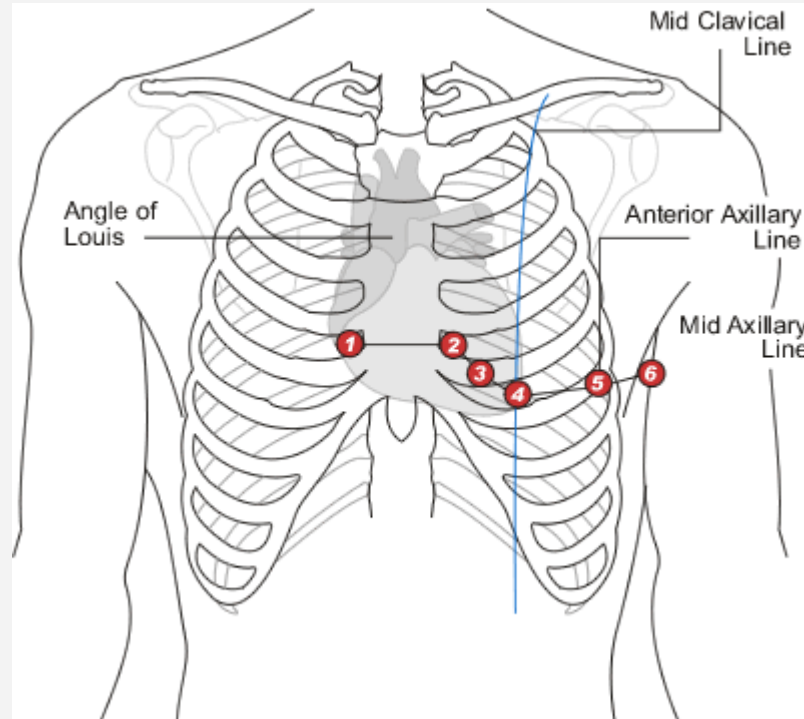
Bike = Blue = V4

On = Orange = V5

Pavement = Purple = V6



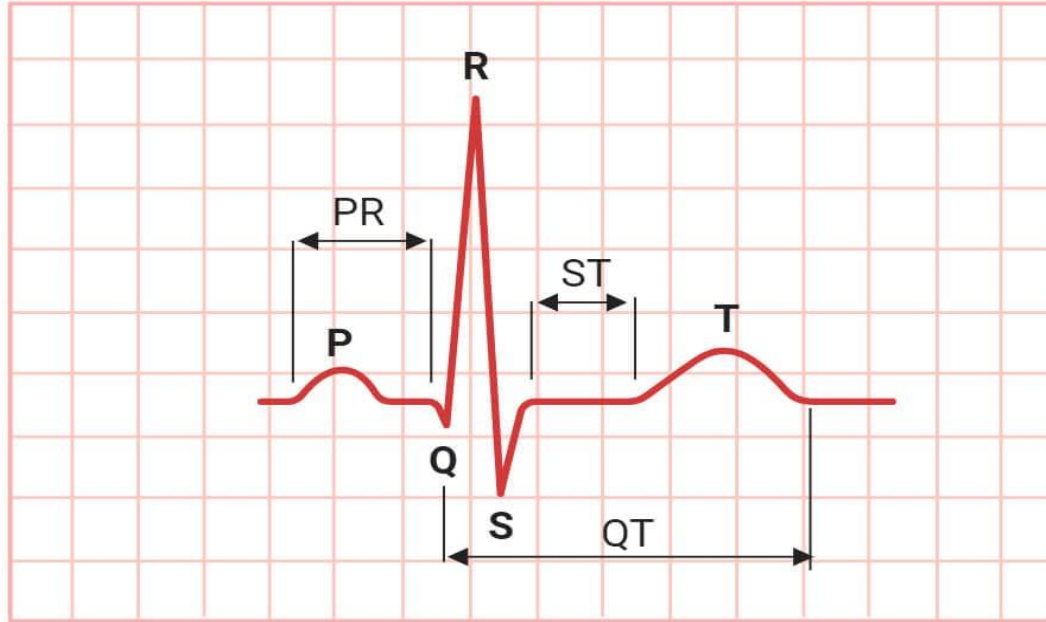
Precordial electrodes



The 10 electrodes in a 12-lead ECG

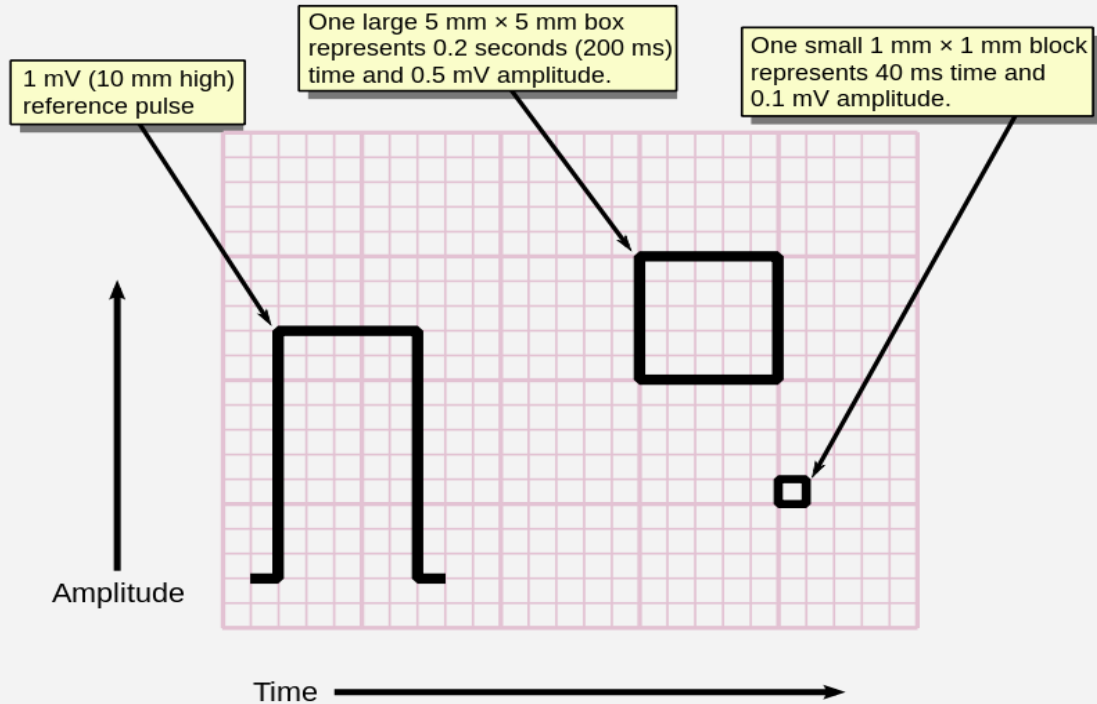
Electrode name	Electrode placement
RA	On the right arm, avoiding thick muscle.
LA	In the same location where RA was placed, but on the left arm.
RL	On the right leg, lower end of inner aspect of calf muscle. (Avoid bony prominences)
LL	In the same location where RL was placed, but on the left leg.
V ₁	In the fourth intercostal space (between ribs 4 and 5) just to the right of the sternum (breastbone)
V ₂	In the fourth intercostal space (between ribs 4 and 5) just to the left of the sternum.
V ₃	Between leads V ₂ and V ₄ .
V ₄	In the fifth intercostal space (between ribs 5 and 6) in the mid-clavicular line.
V ₅	Horizontally even with V ₄ , in the left anterior axillary line.
V ₆	Horizontally even with V ₄ and V ₅ in the mid-axillary line.

EKG/ECG Strip

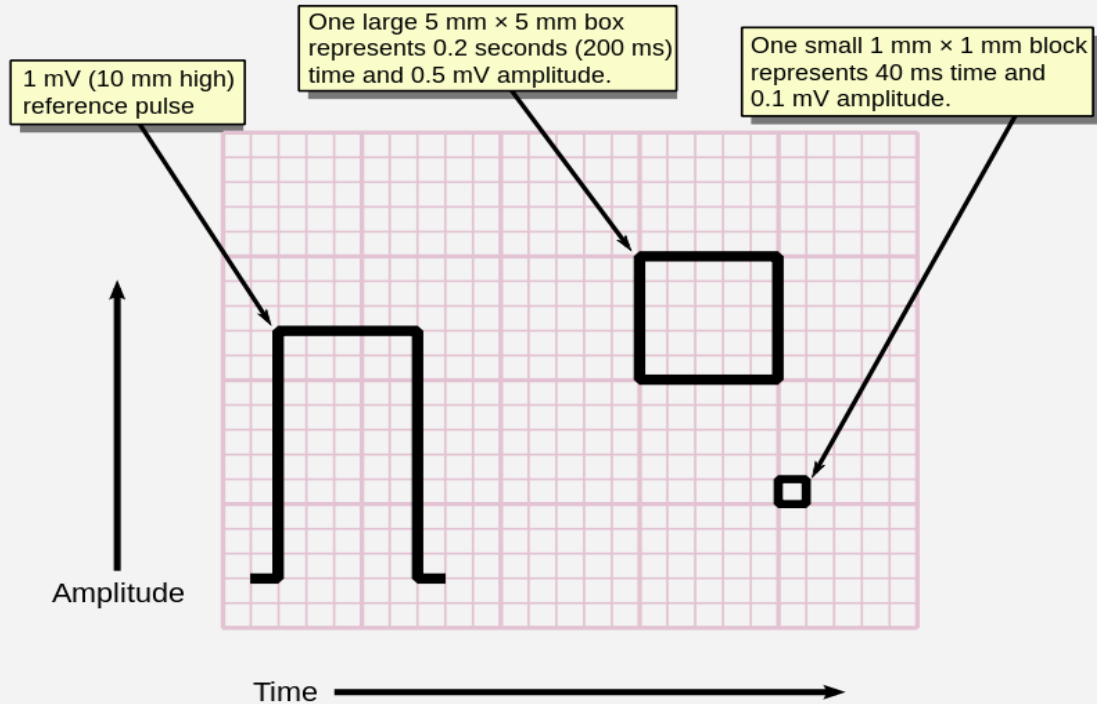


Electrocardiogram
(ECG)

EKG/ECG Strip



EKG/ECG Strip



6-step method ECG/EKG interpretation

- Identify and Examine the P Wave (present and upright) absent or inverted could indicate a form of dysrhythmia
- Measure the PR interval distance from the beginning of p wave and the beginning of QRS complex, count the number of small boxes in between and multiply it by 0.04 sec. Our magic number for the PR interval is 0.12 to 0.20 sec more than 0.20 indicate heart block .
- Measure the QRS complex: Count small boxes in between QRS complex, our magic number is QRS: 0.06 – 0.12 sec
- Identify the rhythm of our strip distance between two QRS complex (two R's)



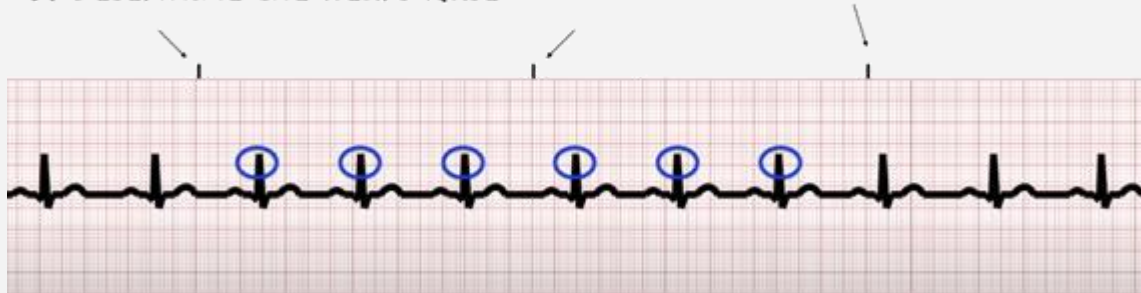
6-step method ECG/EKG interpretation

- Determine the heart rate
 - 6 second method = count the number of R's in between and multiply them by 10 = heart rate
60 beats per minutes , Work best for irregular rhythm



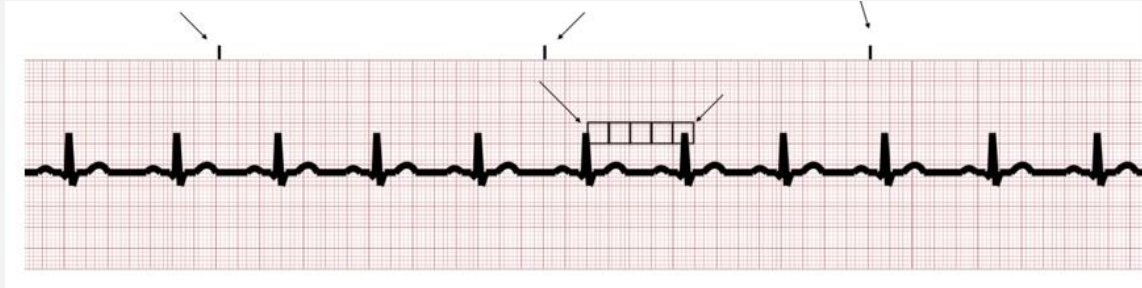
5. Determine the Heart Rate

60 BPM



6-step method ECG/EKG interpretation

- Determine the heart rate:
 - Big box method : 300 divided by the number of big boxes between two R's
 - Formula $300/5=60$ beats per minutes BPM for regular rhythm if you do not have a 6-second strip in front of you.



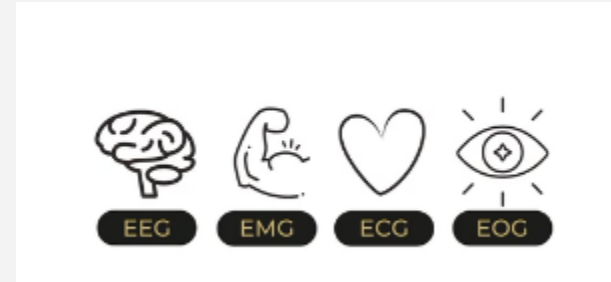
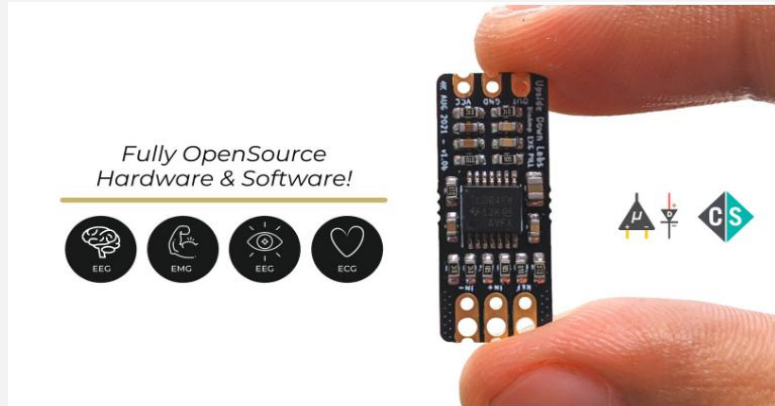
ECG sensors



ECG sensors

BioAmp EXG Pill

Professional-grade analog front-end amplification for ECG, EMG, EOG, and EEG biosensing on one tiny board



ECG Projects

ECG Heartbeat Categorization Dataset

Segmented and Preprocessed ECG Signals for Heartbeat Classification



<https://www.kaggle.com/datasets/shayanfazeli/heartbeat>

ECG Heartbeat Categorization Dataset

Data Card Code (214) Discussion (30)

761

New Notebook

This dataset is composed of two collections of heart-beat signals derived from two famous datasets in heartbeat classification: the MIT-BIH Arrhythmia Dataset and The PTB Diagnostic ECG Database. The number of samples in both collections is large enough for training a deep neural network.

This dataset has been used in exploring heartbeat classification using deep neural network architectures, and observing some of the capabilities of transfer learning on it. The signals correspond to electrocardiogram (ECG) shapes of heartbeats for the normal case and the cases affected by different arrhythmias and myocardial infarction. These signals are preprocessed and segmented, with each segment corresponding to a heartbeat.

Content

Arrhythmia Dataset

- Number of Samples: 109446
- Number of Categories: 5
- Sampling Frequency: 125Hz
- Data Source: Physionet's MIT-BIH Arrhythmia Dataset
- Classes: ['N': 0, 'S': 1, 'V': 2, 'F': 3, 'Q': 4]

The PTB Diagnostic ECG Database

About Dataset

Context

ECG Heartbeat Categorization Dataset



Thanks!
